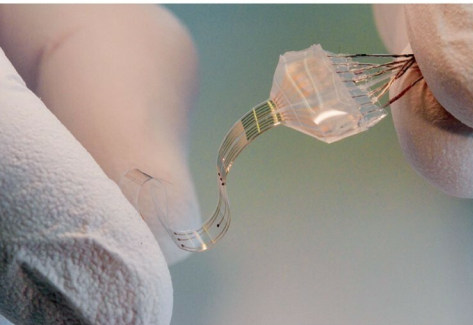




e-Dura implant can help paralysed people to walk again

using carbon nanotubes and sheets of graphene only a few atoms thick.

Helping hand from Big Data

There are many more devices in the wearable cadre; however, the magic ingredient that will make these really useful is smart data analysis.

Dr Michael Docktor, a gastroenterologist at Boston Children's Hospital, also clinical director of innovation and director of clinical mobile solutions, wrote in January in an article about CES 2015, "As these wearables become more of a clinical tool than merely a personal wellness device, data analytics and the ability to serve up meaningful, reliable and actionable data to one's care team will be critical."

Robots to the rescue

Carol Reiley, an experienced computer scientist and roboticist, spoke about future applications of robots in medical technology, at Bay Area Maker Faire 2015. The picture she painted was fantastic—of miniature robots that enter the body through natural openings like the mouth and perform surgeries inside. Some of these are devices like Pillcam that allow minimally-invasive imaging of

a person's colon, sensor-augmented surgical tools like the Intuitive Surgical Firefly system that injects a dye for fluorescent imaging to help identify tumours during surgery, augmented reality solutions that allow surgeons to look beneath the patient's skin, robots that assist in therapeutic exercises and so on.

It is interesting to note that famous robotic contests like John Hopkins Robo Challenge have some quests that encourage children to solve problems similar to those experienced in the medical field. A challenge to build a robot that can spot a grape in Jello, for example, can be compared with finding a tumour in soft tissue.

Implants aplenty

When somebody says implant, you might be immediately reminded of devices like the pacemaker, which have been around for decades now. Now, take a look at these implants and you will realise the current generation is incomparable to medical electronics a decade back.

Dissolvable devices for drug delivery. Last year, researchers at Tufts University's School of Engineering developed resorbable, wirelessly-

controlled electronic implants made of silk and magnesium that could deliver heat treatment on a patient's body tissues to treat bacterial infections and then safely dissolve. The device consisted of a serpentine resistor and a power-receiving coil made of magnesium deposited onto a silk-protein layer. The silk pocket protected the electronics and controlled the device's dissolution time.

In a test on tissue infected with *Staphylococcus aureus* (*S. aureus*), devices were triggered through a wireless transmitter to administer two 10-minute heat treatments. The devices safely dissolved within two weeks. In a subsequent *in-vitro* lab experiment, another device was made to administer antibiotic ampicillin to kill *E. coli* and *S. aureus*.

According to Hu Tao, first author of the study, "The new wireless therapy devices are robust enough to survive mechanical handling during surgery but designed to harmlessly dissolve within minutes or weeks, depending on how the silk protein was processed."

Helping paralysed people walk again. Historically, one of the biggest problems concerning implants has been the mechanical mismatch between soft tissue and stiff implants. For some, this can cause minor issues like inflammation, while for others, it might mean outright rejection by the body.

A team of multi-disciplinary experts in Switzerland has developed a soft, flexible implant called e-Dura, which mimics the shape and elasticity of dura mater, the protective membrane of the brain and spinal cord.

Armed with electrodes, interconnects and chemotrodes that can handle millions of cycles of mechanical stress, chemical injections and electrical stimulation pulses, the device enables those with a paralysed spinal cord to walk again. It is also capable of delivering electrical impulses and chemicals and even monitoring electrical signals from the brain in real-time. e-Dura, which has been tested successfully on